

Multiple Choice

1. **Answer:** A

Options: A) endergonic reaction.; B) catabolic reaction.; C) redox reaction.; D) anabolic reaction.; E) exergonic reaction.

Note: Correct option: A - endergonic reaction.

2. **Answer:** B

Options: A) Simple diffusion can only transport minerals when they are bound to oxygen molecules; B) Simple diffusion cannot occur against a concentration gradient; C) Simple diffusion requires specific carrier proteins for each mineral; D) Simple diffusion only works for organic compounds, not inorganic minerals; E) Simple diffusion can transport minerals in any concentration

Note: Correct option: B - Simple diffusion cannot occur against a concentration gradient

3. **Answer:** A

Options: A) Adenosine triphosphate; B) Nitric oxide; C) Guanosine triphosphate; D) Cyclic GMP; E) Prostaglandins

Note: Correct option: A - Adenosine triphosphate

4. **Answer:** A

Options: A) insulin decreases storage of glycogen; B) growth hormone stimulates cell division and protein synthesis; C) epinephrine increases blood sugar levels and heart rate; D) thyroxine increases the rate of metabolism; E) vasopressin stimulates water reabsorption in the kidney

Note: Correct option: A - insulin decreases storage of glycogen

5. **Answer:** A

Options: A) Prostaglandins are used to enhance mood and have no effect on birth control.; B) Prostaglandins increase fertility by promoting ovulation.; C) Prostaglandins are able to prevent pregnancy by altering the DNA of the sperm and egg.; D) Prostaglandins are known to mediate hormonal action; E) Prostaglandins are non-hormonal, reducing the risk of hormonal side effects.

Note: Correct option: A - Prostaglandins are used to enhance mood and have no effect on birth control.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Extreme diversity, taxonomic organization, possession of certain plant-like features'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Compare gene frequencies and genotype frequencies over two generations'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The vegetal pole undergoes cleavage more readily than does the animal pole'.

Long Form Response

11. **Answer:** Translation. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Electron's energy remains constant but its position changes. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key